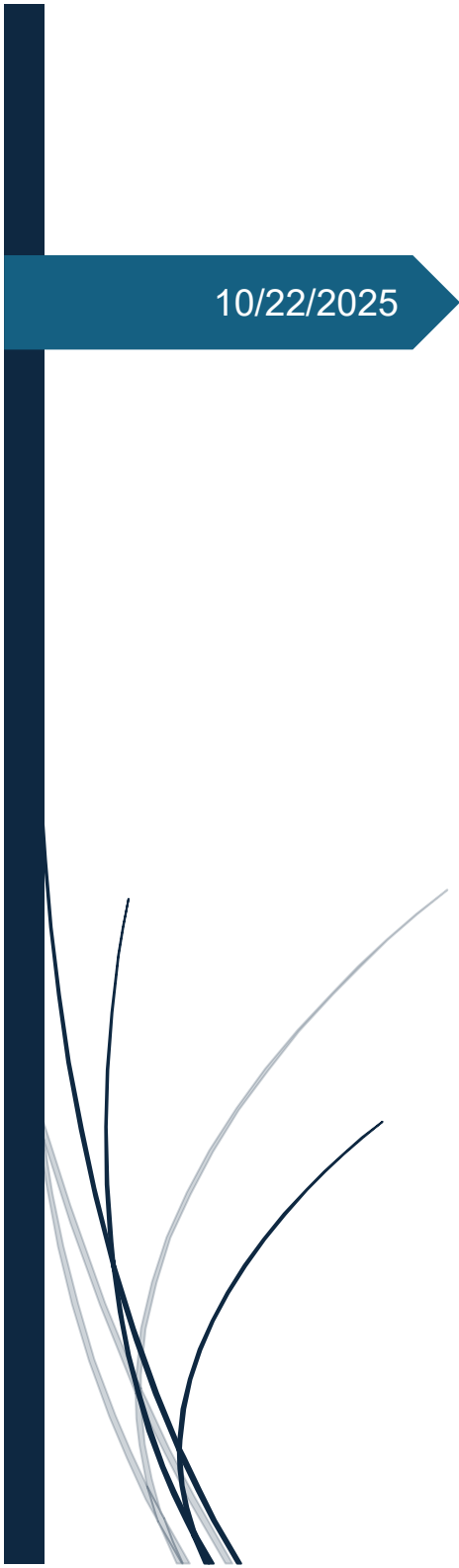


A dark blue vertical bar runs down the left side of the page. A blue arrow points to the right from the bar, containing the date. At the bottom left, there are stylized wheat stalks in dark blue and light grey.

10/22/2025

Leveraging AI for Innovative Wheat Disease and Pest Classification and Treatment Solutions

Using a Convolutional Neural Network to develop a model that accurately classifies both diseases and pests, while also giving suitable treatment options to minimize yield loss caused by diseases.

Prepared by Tam Nguyen

Under the Guidance of Dr. Himanshu Patel

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Saskatchewan Polytechnic



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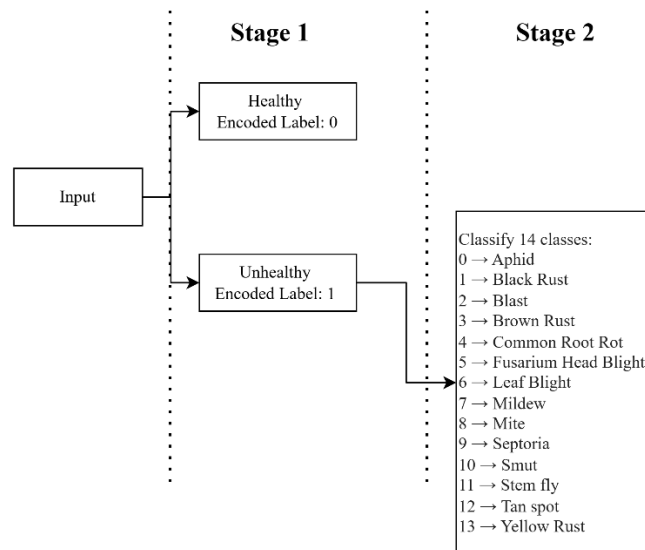
I. Introduction

Wheat is the leading food grain and plays a vital role in reducing global hunger, accounting for 35% of the world's grain supply [1]. It is also Canada's top field crop. Canadian wheat is renowned for its exceptional quality, particularly since Canada produces the most protein-rich winter wheat in the world, specifically Canadian Western Red Spring wheat [2]. The largest single wheat field in the world, recognized by Guinness World Records, is also located in Alberta, Canada [3].

Although Canada produces high-quality wheat and it is the country's leading field crop, it ranks only sixth globally in wheat production. The top five countries each contribute more than 10% to global wheat output, while Canada accounts for only 4% [4]. According to the Crop Protection Network, in 2024, 27% of wheat in Alberta and Ontario was lost to diseases, compared to just 8.3% in U.S. states [5]. Consequently, early identification of plant diseases is vital for increasing crop yields and strengthening agricultural economies.

A. Project Overview

This project aims to utilize Convolutional Neural Networks to develop a model capable of accurately detecting wheat diseases and pest infestations. It will provide treatment recommendations to users or send alerts to a field-quality control system to help reduce yield loss from these issues. The project will have two stages of classification as shown below.



B. Problem Statement

To minimize yield losses due to diseases and pests, timely intervention is crucial. Several studies have utilized computer vision and machine learning for early threat detection, highlighting the importance of identifying and classifying these issues.

Most existing studies primarily focus on classifying either diseases or pests individually, and their scope remains limited. Additionally, most studies stop at detection and classification, without developing methods for treating or preventing the identified threats.

This project aims to overcome the mentioned limitations by developing a deep learning model that can accurately identify both diseases and pests, while also providing suitable treatment options.

C. Research Questions/Hypotheses

Below are three research questions and three hypotheses:

	Key Aspects	Research Questions	Hypotheses
1	Dataset requirements	How do imbalanced classes affect the model's accuracy, and what techniques can be used to mitigate their effect on the model?	Utilizing techniques to mitigate the effects of imbalanced classes can enhance the model's accuracy.
2	Model selection	Is a single-stage classification (classifying all classes at once) more accurate, or is a two-stage classification (first identifying diseases and pests, then determining the specific type of disease or insect) more accurate?	A two-stage classification pipeline will typically yield higher accuracy than a one-stage approach.
3	Model accuracy	Can an AI model accurately identify wheat disease types and pest infestations?	Using a deep learning model to classify can bring a high level of accuracy

II. Literature Review

Two studies and literature reviews will be discussed in this section. Below are summaries of both studies.

Literature name	Study for Aphid Cluster Detection and Segmentation in Sorghum Fields	Recognition of Wheat Leaf Diseases Using Lightweight Convolutional Neural Networks against Complex Backgrounds
Purpose	Classify and detect if Sorghum fields have aphid clusters or not.	Recognize wheat stripe rust and wheat powdery mildew
Data collection	5,447 images collected from actual sorghum fields	1,350 images with two disease classes: wheat stripe rust and wheat powdery mildew

Literature name	Study for Aphid Cluster Detection and Segmentation in Sorghum Fields	Recognition of Wheat Leaf Diseases Using Lightweight Convolutional Neural Networks against Complex Backgrounds
Training Method	● Pre-processing: Resized and Normalized ● Implement 10-fold-cross validation ● Employ patch generation with three different scales ● Deploy different learning strategies to different models	● Pre-processing: Resized and Normalized ● Apply data augmentation ● Optimize model by employing different training strategies, with three initial learning rates (0.05, 0.01, and 0.001)
Used Models	● Four segmentation models: Fast-SCNN, BiSeNetV1, BiSeNetV2, and Small HRNet ● Three object detection models: Faster R-CNN, RT-DETR, and YOLOv7	Five lightweight convolutional neural network models: MobileNetV3, ShuffleNetV2, GhostNet, MnasNet, and EfficientNetV2
Evaluation	● The segmentation models: mean Average Precision (mAP) and Intersection over Union (IoU). ● The detection models: Intersection over Union and Dice coefficient. ● The speed of all models is evaluated based on how fast they can detect aphid clusters by their frames per second (FPS)	● Accuracy ● Precision ● Recall ● F1 score ● The confusion matrix heat map
Result	● Top performance segmentation model: Fast-SCNN with the highest real-time speed at 91.66 FPS and a high accuracy level of 71.25 mIoU. ● Top performance detection model: RT-DETR with the highest accuracy with 61.63 mAP and the second fastest model at 72.55 FPS.	The MnasNet model, that used both SGD + StepLR with the initial learning rate of 0.001, had the highest recognition accuracy of 98.65%

Key findings

- It is important to minimize the impact of an imbalanced dataset by using class weights to help the model learn more effectively.
- Detecting insect clusters is more accurate than identifying individual insects because clusters are easier to detect.

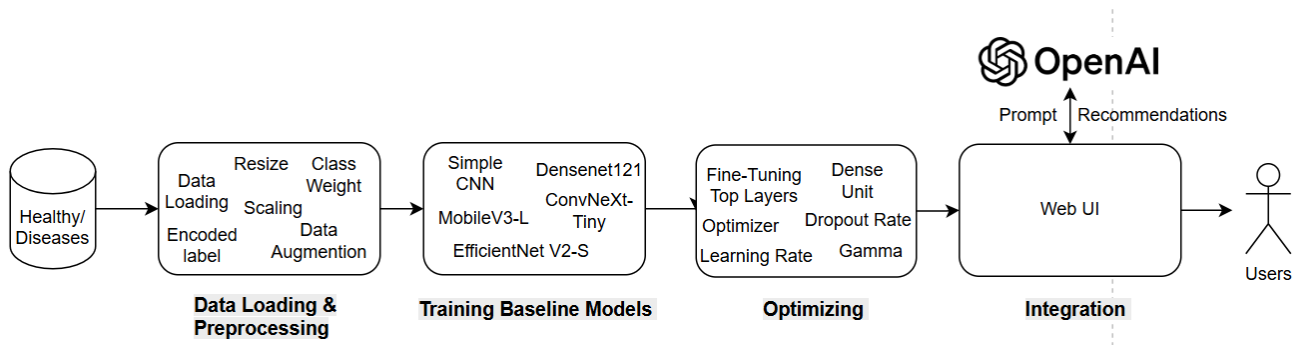
- The experiment results clearly show the trade-off between speed and accuracy; achieving higher accuracy takes more time. The best model is one that finds a good balance among the key metrics listed. However, it is also important to prioritize these key metrics.
- The choice of learning strategies and hyperparameters significantly influences the model's performance. To achieve optimal results, it is advisable to experiment with various learning strategies and initial learning rates for each model.

Gap

- Since the images used in “Study for Aphid Cluster Detection and Segmentation in Sorghum Fields” were captured on sorghum plants, not wheat, the leaf structure and pest behavior might differ.
- These studies do not cover a broad range of pests and diseases for classification.
- These studies do not include any user interface for farmers. It focuses on segmentation and detection but does not extend to real-world applications.

III. Methodology

The diagram below shows the main stages of this project.



A. Data Collection

Dataset explanations: This dataset includes 13,104 high-resolution images of real-world wheat diseases, excluding artificially augmented images.

Dataset link: <https://www.kaggle.com/datasets/kushagra3204/wheat-plant-diseases>

- Number of images: 13,104 high-resolution images
- Number of classes: 15 Classes (1 healthy class, pest: 3 classes, leaf/root diseases: 11 classes)

Sample images from the dataset with their corresponding ground truth labels.



				
Stripe Rust	Stem Rust	Common Root Rot	Leaf Blight	Wheat Blast
				
Fusarium head blight	Septoria Leaf Blotch	Smut	Tan Spot	Powdery Mildew

Table 1. Sample Images from The Dataset

B. Detailed Project Workflow

The following outlines each project stage in detail, from data collection to user interface development.

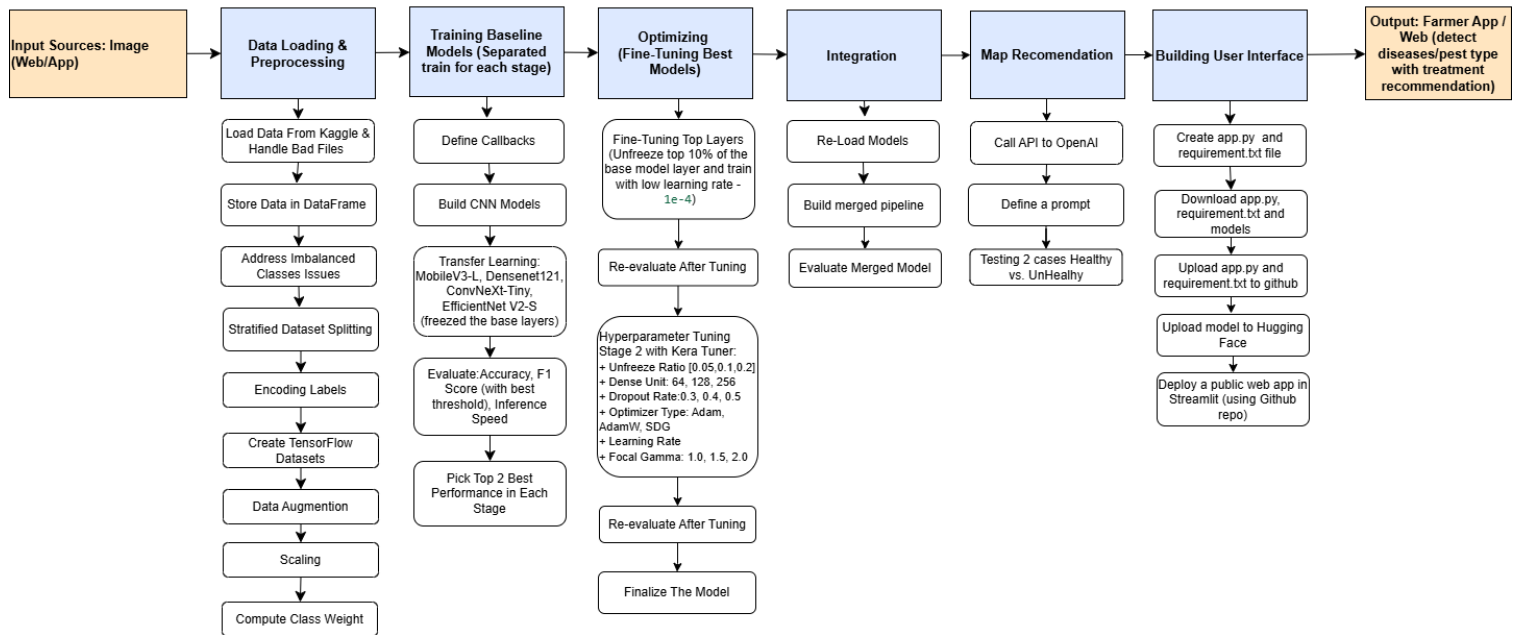


Table 2. Project Workflow Diagram

C. Project Approaches and Clarification

Approach 1: Building A Two-Stage Classification Model

Problem: The “Healthy” class has more images than the 9 disease classes. When using a one-stage classification that predicts all 15 classes simultaneously, the model might be biased toward the majority class, such as "Healthy," leading it to miss actual disease cases.

Class Names	Data Size	Percentage(%)	Rank
Smut	1,310	9.997	1
Yellow Rust	1,301	9.928	2
Brown Rust	1,271	9.699	3
Septoria	1,144	8.730	4
Mildew	1,081	8.249	5
Healthy	1,000	7.631	6
Aphid	903	6.891	7
Leaf Blight	842	6.426	8
Mite	800	6.105	9
Tan spot	770	5.876	10
Blast	647	4.937	11
Common Root Rot	614	4.686	12
Fusarium Head Blight	611	4.663	13
Black Rust	576	4.396	14
Stem fly	234	1.786	15
Total	13,104		120

Table 3. Class Distribution Table

To test the above hypothesis, a CNN model that classify all 15 classes at once with stratified splitting, data augmentation, and class weight was used, achieving only 61% accuracy, a loss of 1.35, and a F1 score of 0.07. In addition, the confusion matrix indicates that the model overpredicts the “Healthy” class.

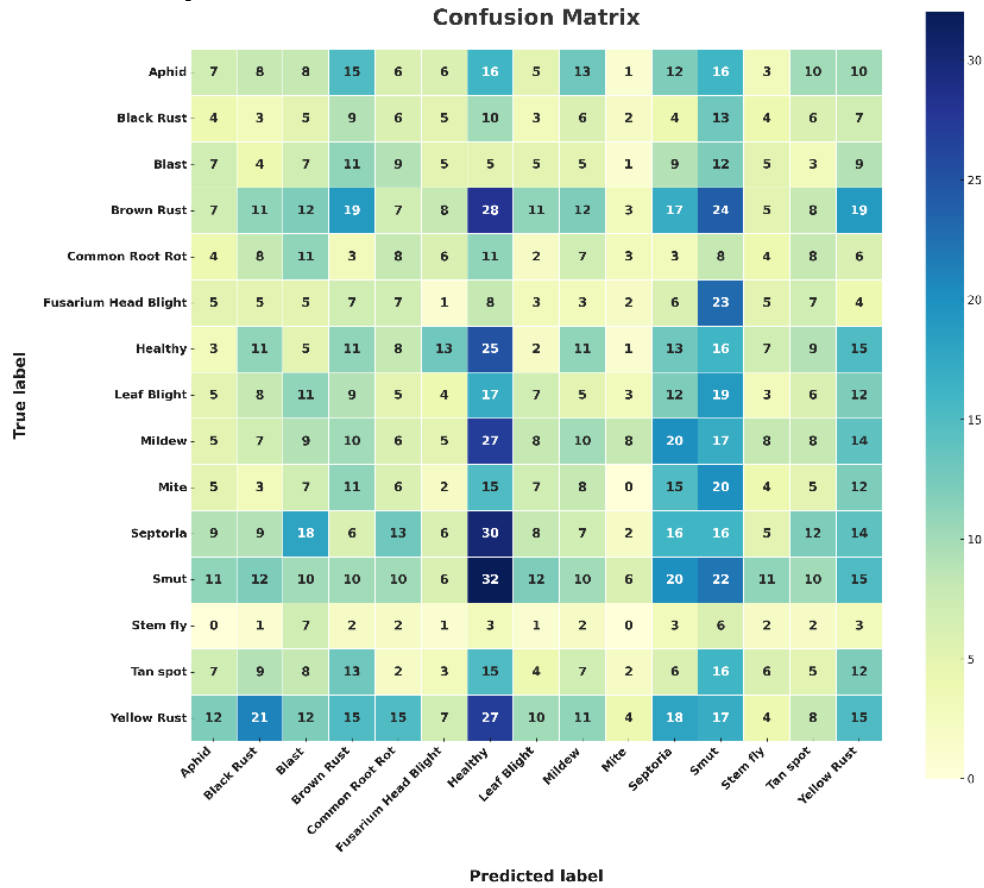


Table 4. The Confusion Matrix of The One-Stage Classification Model

Solution: In this project, a two-stage classification will be used. The first stage is binary classification (Healthy vs. Unhealthy). In this stage, the model can easily identify an unhealthy plant by detecting general damage cues or discoloration. If the plant is identified as unhealthy, it will proceed to the second stage, which classifies 14 types of diseases and pests. By filtering out the “Healthy” class, the model can focus more to learn finer disease-specific differences.

Approach 2: Using Stratified Splitting and Class Weight to Mitigate Imbalanced Classes

Problem: The dataset is highly imbalanced. The class with the most images has five times as many images as the class with the fewest. Without any treatments, the model will tend to produce more images of the majority class when splitting the dataset, resulting in bias toward it.

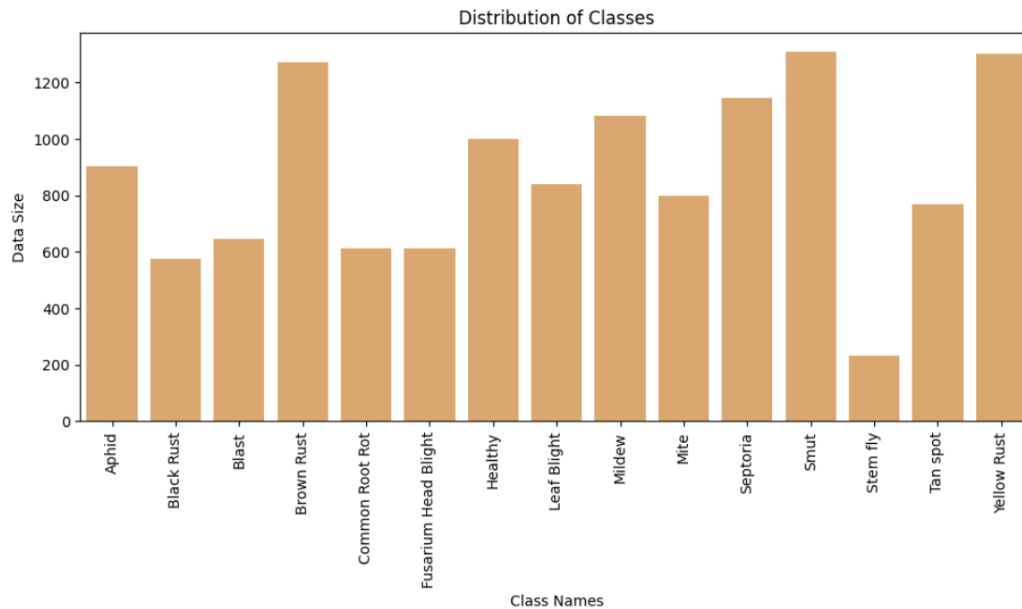


Table 5. Visualization of Class Distribution

Solution: Using stratified splitting to maintain the ratio of each class when dividing into train, test, and validation sets. Additionally, computing class weights with the 'balanced' method ensures that classes with more images receive lower weights. During training, the loss is multiplied by the class's weight, allowing the model to focus more on rare classes.

	Label_Name	Total_Images	Weight
0	Stem fly	164	3.689895
1	Black Rust	403	1.501595
2	Fusarium Head Blight	428	1.413885
3	Common Root Rot	430	1.407309
4	Blast	453	1.335856
5	Tan spot	539	1.122714
6	Mite	560	1.080612
7	Leaf Blight	589	1.027407
8	Aphid	632	0.957505
9	Mildew	757	0.799396
10	Septoria	801	0.755484
11	Brown Rust	889	0.680701
12	Yellow Rust	910	0.664992
13	Smut	917	0.659916

Table 6. Final Class Weight

Approach 3: Utilizing Data Augmentation to Increase the Diversity

Problem: The dataset includes many diseases with similar features, such as Brown Rust and Yellow Rust, which share overlapping textures, colors, and leaf pattern characteristics. This could make it harder for the model to distinguish between similar diseases.



Black Rust Image



Yellow Rust Image

Solution: Use global data augmentation on all classes (flip, rotate, zoom in and out, adjust contrast, brightness, and shift horizontally and vertically) to increase dataset diversity.



Table 7. Sample Images after Data Augmentation

Approach 4: Using Transfer Learning to Improve Accuracy

Problem: The baseline CNN model trained from scratch achieves low accuracy, especially in stage 2 classification, with only 55.4%. According to the literature review, pre-trained models generally achieve higher accuracy because they are pre-trained on large, diverse datasets. The table below compares 10 common pre-trained models.

#	Model	ImageNet Top-1 Accuracy (%)	Parameters (Millions)	GFLOPs (Approx.)	Model Size (MB)	Web/App Friendly Rating	Publication Year
1	EfficientNet-V2-S	87.3	22	8.4	85	Medium	2021
2	ConvNeXt-Tiny	87.8	28.6	4.5	110	Medium	2022
3	MobileNetV3-Large	75.2	5.4	0.22	22	Very Fast	2019
4	DenseNet-121	79.6	8	2.9	33	Medium	2016
5	EfficientNet-B0	77.1	5.3	0.39	20	Fast	2019
6	ResNet-50	76	25.6	4.1	98	Medium	2015
7	VGG-16	73.2	138	15.5	528	Very Slow	2014
8	MobileNetV2	72	3.5	0.3	14	Fast	2018
9	SqueezeNet	57.2	1.3	0.3	5	Very Fast	2016
10	AlexNet	60.3	60	0.72	240	Slow	2012

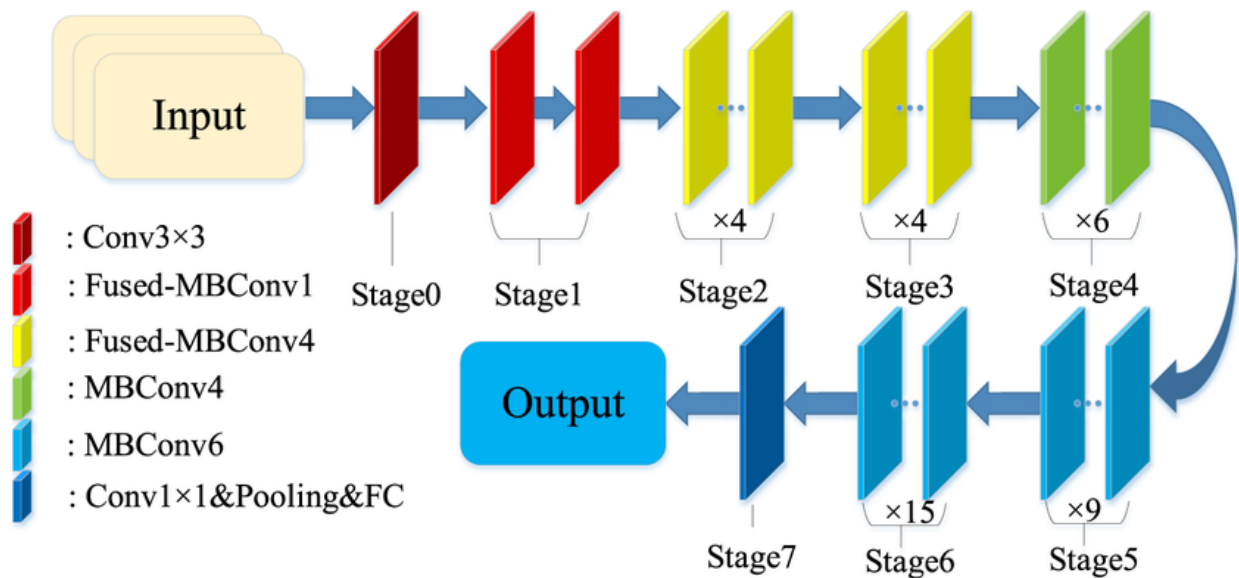
Table 8. Comparison of the Top 10 Pre-trained Models [6-14]

Solution: Based on the comparison above, considering accuracy and web/mobile friendliness, this project will use EfficientNet V2-S, ConvNeXt-Tiny, MobileNetV3-Large, and DenseNet-121. All pre-trained models will start with their layers frozen. Then, the top two models in each stage will have some of their top layers unfrozen to better adapt to the dataset.

D. Architecture of Pretrained Model

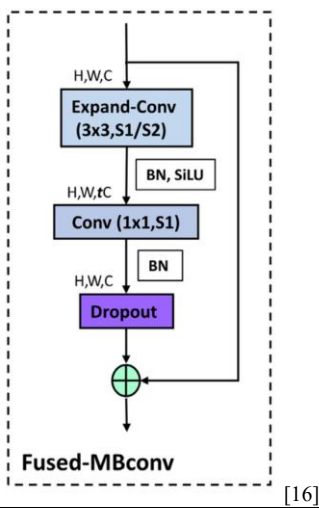
1. Architecture of EfficientNet V2-S

EfficientNetV2-S features multiple stages with repeated convolutional blocks. The Fused-MBConv accelerates early feature extraction, boosting efficiency, while deeper MBConv layers with Squeeze-Excitation improve accuracy and capture complex textures. The high-level architecture of EfficientNetV2-S is shown below.



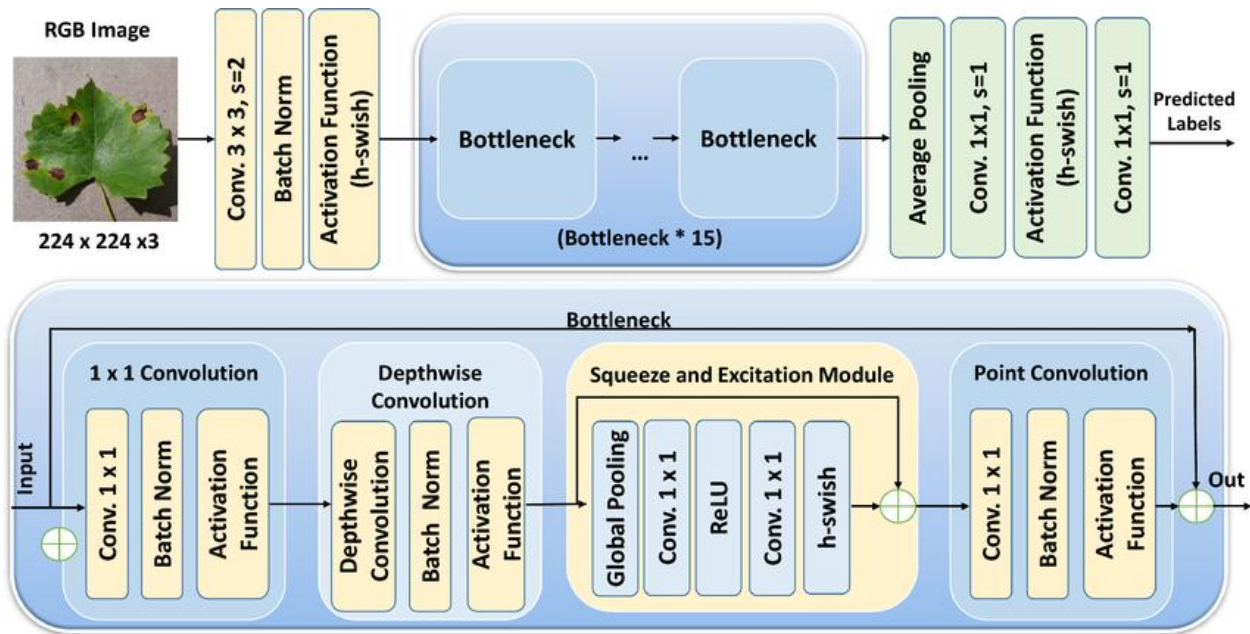
The EfficientNetV2-S is capable of efficient fine-grained classification, offering a good balance of speed and accuracy. It is well-suited for this project. The table below explains the role of each EfficientNetV2-S block in this project.

Stage	Block Type	Channels	Blocks	Stride	Feature map	Functions
Stem	Conv 3×3	24	1	2	224 → 112 x 112	- Extracts basic edges and textures - Prepares the first low-level feature map so deeper layers can learn useful patterns

Stage	Block Type	Channels	Blocks	Stride	Feature map	Functions
Stage 1		24 → 48 → 64	2	1, 2	112 x 112 → 56 x 56	Capture simple textures, leaf colors, and common disease patterns.
Stage 2		64 → 80	4	2	56 x 56 → 28 x 28	Begin capturing more detailed features, such as disease patch shapes.
Stage 3		80 → 160	6	2	28 x 28 → 14 x 14	Understands medium-sized structures, such as the structure of leaf veins, shapes of disease clusters, and fungal patterns
Stage 4		160 → 256	9	2	14 x 14 → 7 x 7	Features extracted fine-grained texture differences, subtle contrast differences, directional marks, and fungal patterns.
Stage 5		256 → 512	15	1	7 x 7	• Learn high-level disease representations • Distinguish look-alike diseases
Stage 6		256 → 512	15	1	7 x 7	• The brain of the model • Combines all patterns learned previously • Integrates both fine textures and global patterns
Stage 7	Conv → BatchNorm → SiLU → Global Average Pool → Dropout → Fully Connected	1280	1	—	—	• Improves generalization • Produce final disease class probabilities.

2. Architecture of MobileNetV3-Large

MobileNetV3-Large is a highly efficient, lightweight CNN that balances computational speed with high classification accuracy. The high-level structure of MobileNetV3-Large is illustrated below.



High-level architecture of MobileNetV3-Large [17]

MobileNetV3-Large is significantly smaller and faster than EfficientNetV2-S, with 24 million parameters compared to 5.4 million. It uses the Inverted Residual Block design from MobileNetV2, the Hard-Swish activation, and selective SE attention to effectively capture important features at lower computational cost. Its design, optimized through Neural Architecture Search, balances block types, expansions, kernel sizes, and activations for optimal performance. As a result, MobileNetV3-Large is ideal for real-time applications and lightweight deployment.

E. Evaluation

In this project, all models will be evaluated based on **accuracy, F1 score, and inference speed.**

- **Accuracy**
 - Measures how correctly the model classifies images, calculated by the ratio of correct predictions over total predictions.
 - It is an important metric for assessing general performance in both stages
- **F1 Score**
 - Measures based on a combination of Precision and Recall.
 - Formulation: $F1 \text{ Score} = 2 \times (\text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall}))$
 - It's important when evaluating an imbalanced dataset. By monitoring F1 score, we ensure that the model performs well not just on common classes but also on rare cases
 - For this project, the primary evaluation metric for the top-performing model will be the F1 Score.
- **Inference Speed**
 - Evaluate how fast a model can process input images and generate predictions
 - Formulation:
 - $\text{Images per sec} = \text{Number of Images} / \text{Total Time} \rightarrow \text{How many images the model can process per second}$
 - $\text{Average time per image in milliseconds (ms per Image)} = (\text{Total Time} / \text{Number of Images}) \times 1000$

- This metric is crucial for real-time applications like web or mobile deployment. Typically, it involves a trade-off between speed and accuracy. However, in this project, accuracy and the F1 score are more significant.

F. Ethical Considerations

There are two ethical considerations for this project:

- **Data privacy and consent:** Since this dataset was not collected directly and consent was not obtained from the individuals who took the images, it is intended solely for research purposes and cannot be used to inform further actions. For practical applications, obtaining consent from the data source will be essential.
- **Responsible Recommendation:** The system should base its advice on trusted agricultural sources for treatment guidance. Providing vague or overly general advice may lead to crop damage, financial loss, or environmental problems if not followed properly.

IV. Results and Analysis

A. Model Evaluation

1. Stage 1 (Healthy vs. Unhealthy) Result

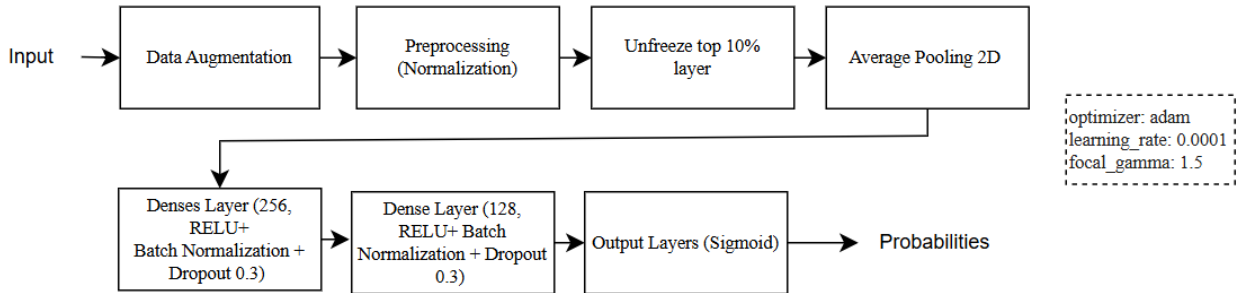
- The final model used for stage 1 classification is EfficientNet V2-S.

Stage 1 - All Models (Before + After Fine-tuning):

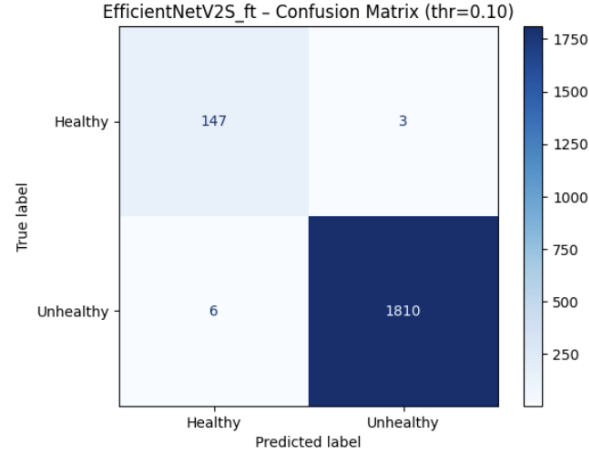
	Model	Accuracy	Loss	Best F1	Best Threshold	Images/sec	ms per image
0	EfficientNetV2S_ft	0.9908	0.0309	0.9975	0.10	128.14	7.80
1	MobileNetV3Large_ft	0.9893	0.0349	0.9961	0.22	220.15	4.54
2	MobileNetV3Large	0.9471	0.1414	0.9884	0.10	238.84	4.19
3	DenseNet121	0.9471	0.1385	0.9871	0.11	142.49	7.02
4	ConvNeXTTiny	0.9547	0.1366	0.9859	0.10	73.65	13.58
5	EfficientNetV2S	0.9018	0.2445	0.9830	0.10	147.26	6.79
6	CNN_Baseline	0.9517	0.1898	0.9767	0.64	330.18	3.03

Table 9. Comparison of All Models in Stage 1 (Before and After Tuning)

- **Model architecture:**



- **Evaluation: The model achieved 99.08% accuracy and a 99.75% F1 score.** Below is the confusion matrix.



2. Stage 2 (14 diseases/pest types) Result

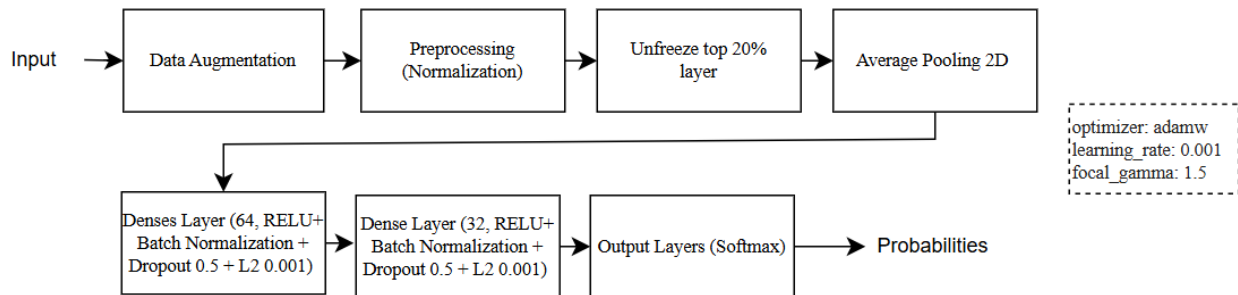
- The final model used for stage 2 classification is also EfficientNet V2-S.

Stage 2 - All Models (Before + After Fine-tuning):

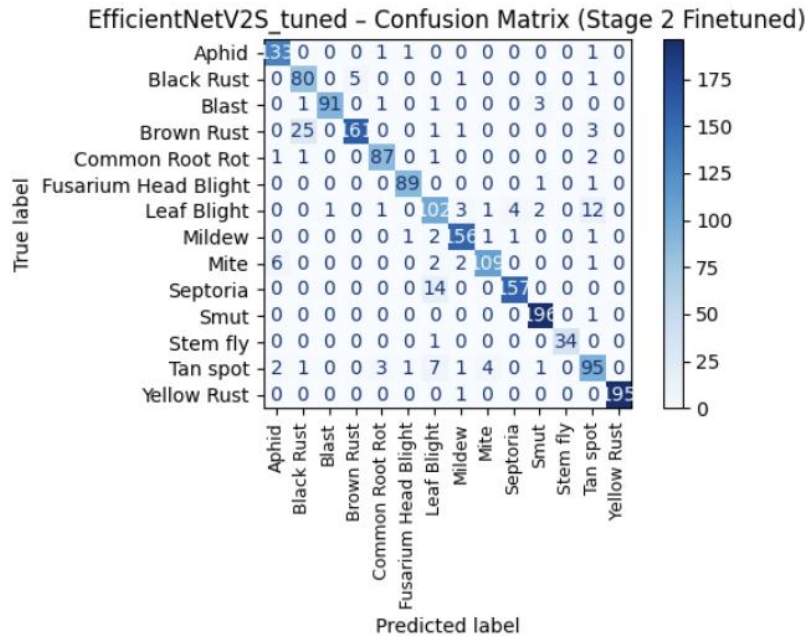
	Model	Accuracy	Loss	F1 (macro)	F1 (weighted)	Images/sec	ms per image
0	EfficientNetV2S_tuned	0.9279	0.0636	0.9252	0.9285	139.17	7.19
1	EfficientNetV2S_ft	0.8805	0.0548	0.8804	0.8830	146.39	6.83
2	MobileNetV3Large_ft	0.8420	0.0745	0.8451	0.8478	209.59	4.77
3	DenseNet121_ft	0.8309	0.0788	0.8329	0.8371	146.81	6.81
4	EfficientNetV2S	0.7941	0.0970	0.7850	0.7956	139.18	7.18
5	MobileNetV3Large	0.7847	0.1096	0.7711	0.7887	214.67	4.66
6	ConvNeXtTiny	0.7819	0.1088	0.7677	0.7833	71.77	13.93
7	DenseNet121	0.7704	0.1135	0.7628	0.7758	143.96	6.95
8	CNN_Baseline	0.5898	0.2501	0.5546	0.5951	305.43	3.27

Table 10. Comparison of All Models in Stage 2 (Before and After Tuning)

- Model architecture:



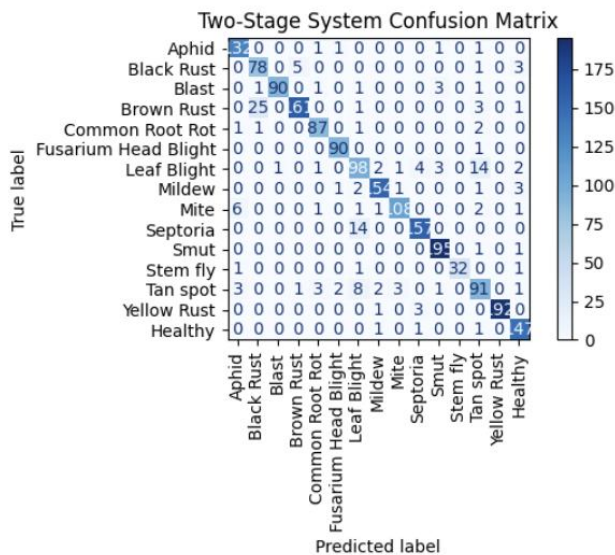
- Evaluation: The model achieved 92.79 % accuracy and a 92.52% F1 score. Below is the confusion matrix.



3. Result after merging two models

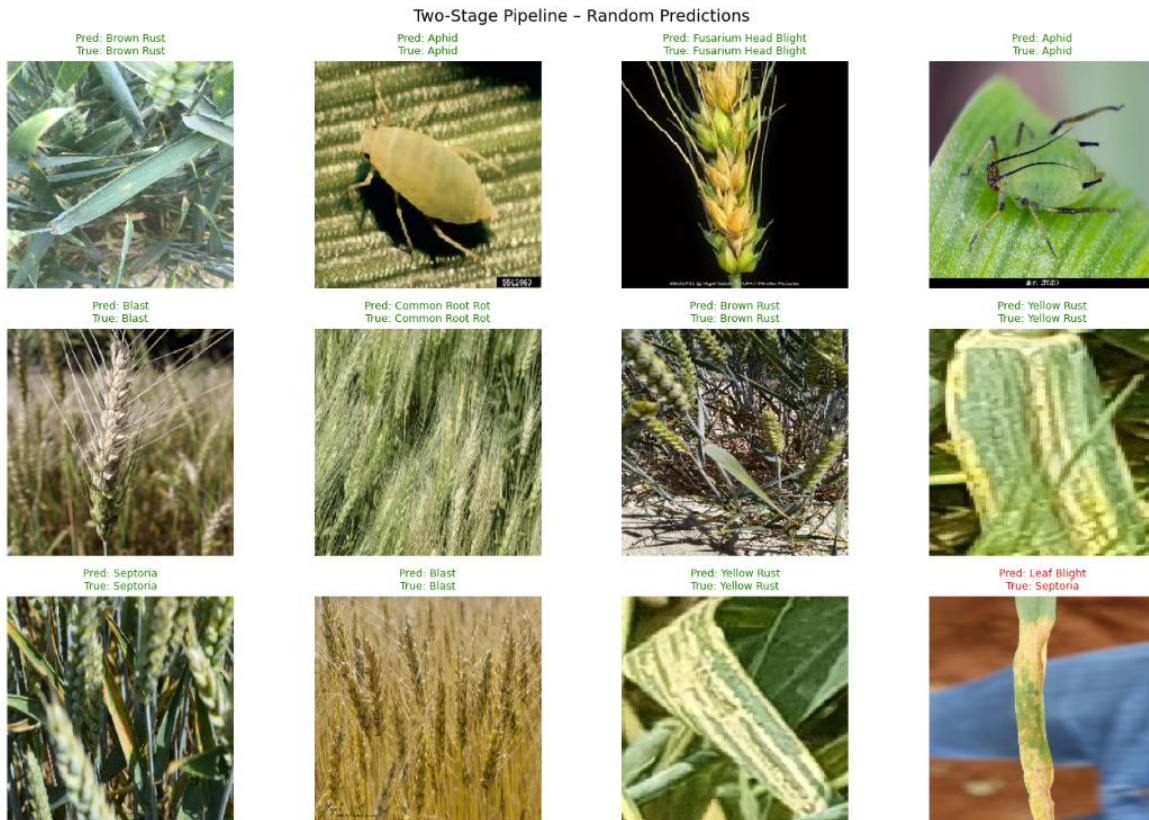
After merging two models, the final model achieved 92.17 % accuracy and a 91.73% F1 score.

Below are the confusion matrix, classification report, and an image showing 12 random predictions.



Classification Report:

	precision	recall	f1-score	support
Aphid	0.923	0.971	0.946	136
Black Rust	0.743	0.897	0.812	87
Blast	0.989	0.928	0.957	97
Brown Rust	0.964	0.843	0.899	191
Common Root Rot	0.926	0.946	0.935	92
Fusarium Head Blight	0.957	0.989	0.973	91
Healthy	0.919	0.980	0.948	150
Leaf Blight	0.772	0.778	0.775	126
Mildew	0.957	0.951	0.954	162
Mite	0.956	0.900	0.927	120
Septoria	0.952	0.918	0.935	171
Smut	0.961	0.990	0.975	197
Stem fly	1.000	0.914	0.955	35
Tan spot	0.765	0.791	0.778	115
Yellow Rust	1.000	0.980	0.990	196
accuracy			0.922	1966
macro avg	0.919	0.918	0.917	1966
weighted avg	0.925	0.922	0.922	1966



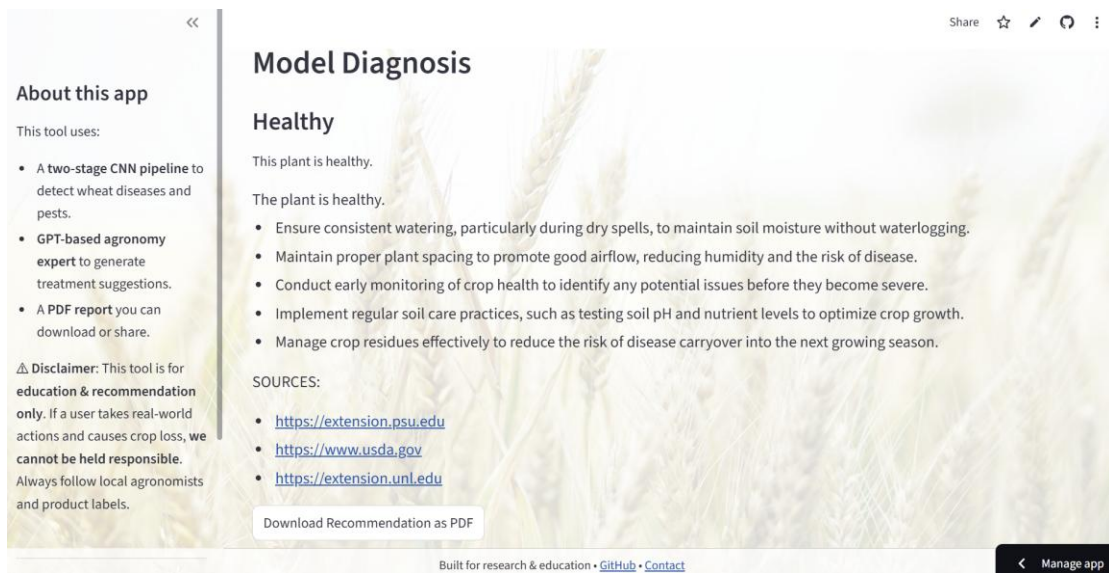
B. Deployment

For deployment, the app.py file has been uploaded to GitHub, and the models are stored on Hugging Face. These are then connected with Streamlit to build the website.

Website link: <https://wheat-disease-detection-app-pxkrzub9lclpqv4qlmhgce.streamlit.app>



Picture 1: Screenshot of the main page (before uploading the image)



Picture 2: Screenshot of the main page (after uploading the image) - Testing healthy case



Picture 3: Screenshot of the main page (after uploading the image) - Testing unhealthy case

V. Budget and Resources

Powerful GPUs are essential for managing large datasets and accelerating training. Entry-level GPUs usually cost between \$250 and \$500. To obtain faster GPUs at a lower price, using Google Collab Pro is the optimal choice for this project, priced at 13.99 CAD per month. The overall estimated cost is approximately 28 CAD.

VI. Conclusion

A. Benefit of the project

This project aims to develop an AI-powered system that detects wheat diseases and pests, providing appropriate treatments to enhance wheat yield. Implementing this model in real-world

settings provides clear benefits at both the microeconomic and macroeconomic levels, as outlined below:

Microeconomic impact:

- Early detections result in healthier crops and increased yields. The sooner the intervention, the more time and opportunity farmers have to act.
- By accurately detecting the right diseases, farmers can select the most effective treatments. This helps farmers reduce costs by avoiding incorrect treatments.

Macroeconomic impacts:

- Increasing the wheat supply helps ensure global food security.
- Having the correct treatment knowledge makes agriculture more sustainable. Instead of applying chemicals indiscriminately, farmers can reduce their chemical usage to decrease pollution.
- Reducing crop losses boosts productivity, driving economic growth by increasing farmers' incomes.

B. Scope of Future Work

- Enhancing diversity by including various dataset types and sizes, such as aerial images and videos.)
- Launching an app that users can access on their smartphones.
- Increasing initiatives to diagnose diseases in a broader range of crops.
- Expanding the website's language options.

VII. References

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